

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q24-2455

Emory Grove 138 kV
November 29, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2455. The Interconnection Customer proposes a 135.0 MW battery energy storage facility interconnecting at the Emory Grove 138 kV substation in the Maryland area. The request was received on September 21, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 416 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$17,452,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Emory Grove – Montgomery Village Substation 138kV	Ritzville – Germantown 138kV	128.9	177	100.0
Middlebrook Substation – Rancho 138kV	Artesia Country Club – Air Force Base 345kV	118.4	320	13.7
Rancho – Air Force Base 138kV	Artesia Country Club – Air Force Base 345kV	116.7	301	13.7
Clover Vly – Ross Fork 138kV line	Artesia Country Club – Air Force Base 345kV	110.3	424	10.5
East Devon – Thiele Kaoli Reedy Crk 345kV Line	Chambersburg 500-345 kV TX	108.0	1,298	6.1
Clover Vly – Middlebrook 138kV	Artesia Country Club Substation – Air Force Base 345kV	102.2	341	11.9
Concord – Bluebonet 230kV	Pueblo Xtra – Boca Raton 500kV	102.2	2,215	6.5
Chambersburg 500-345 kV Transformer	E Dvon – Carrel 345kV	101.1	1,188	6.7
Phillips Pump 1 Substation – Rockville Substation 138kV	Tiger Bay – Air Force Base 345kV Line	100.6	279	25.0
Germantown – Ritzville 138kV	Air Force Base 345 kV/138 kV XFMR	100.5	237	26.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Montgomery Village – Emory Grove 138kV	Rebuild line with higher-capacity conductor	\$1,714,000	100% (\$1,714,000)
Middlebrook Substation – Rancho 138kV	Rebuild line with higher-capacity conductor	\$2,328,000	100% (\$2,328,000)
Rancho – Air Force Base 138kV	Rebuild line with higher-capacity conductor	\$1,031,000	100% (\$1,031,000)
Clover Valley – Ross Fork 138kV Line	Rebuild line with higher-capacity conductor	\$2,723,000	69% (\$1,870,000)
East Devon – Thiele Kaoli Reedy Crk 345kV Line	Reconductor line	\$10,665,000	8% (\$816,000)
Middlebrook – Clover Vly 138kV Line	Reconductor line	\$4,898,000	100% (\$4,898,000)
Concord Substation – Bluebonnet 230kV	Reconductor line	\$7,127,000	1% (\$71,000)
Chambersburg 500-345 kV Transformer	Replace transformer with higher-rated unit	\$56,990,000	4% (\$1,971,000)
Phillips Pump 1 Substation – Rockville Substation 138kV	Reconductor line	\$526,000	70% (\$368,000)
Germantown – Ritzville 138kV Line	Reconductor line	\$2,890,000	82% (\$2,385,000)
	Total allocated to project		\$17,452,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.